Both N=96 levers are real,

and they partly substitute

Stacking cosine lr onto the 80,000-instance recipe gives a new best at N=96:
0.9453 [0.9326, 0.9557], up +0.0180 on the same pool, 88-61, p = 0.033.
But cosine was worth +0.0473 at 20,000 and only +0.0180 at 80,000.

The 2x2 over {20000, 80000} x {constant, cosine} is now complete on one shared pool of
1500 held-out instances, so all four cells are paired-testable against each other.
All four edges are significant; the two gains do not add.

What the model actually sees

Every token is one categorical id into a 106-symbol vocabulary (N=96, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-105	N0 N1 ... N95	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 777 TOKENS (seed 50000000002)

HINTS	BEGIN_HINTS N2 COLOR_1 N3 COLOR_2 N4 COLOR_1 N5 COLOR_3 N6 COLOR_1 N7 COLOR_2 ... (129 tokens)
GRAPH	BEGIN_GRAPH N0 N67 N84 N34 N49 N59 N1 N86 N4 N17 N53 N44 ... (577 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_2 N11 COLOR_2 N16 COLOR_0 N20 COLOR_0 N35 COLOR_1 N37 ... (71 tokens)

Loss is on the REASONING section only: 5 rounds, 32 deductions, 64 target tokens. Everything before BEGIN_REASONING is the problem statement.

One real reasoning trace

64 hints of 96 nodes, then 5 rounds of simultaneous deduction. Hints: N2=1 N3=2 N4=1 N5=3 N6=1 N7=2 N8=3 N9=3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N0 COLOR_2 N11 COLOR_2 N16 COLOR_0 N20 COLOR_0 ...
  18 deduced
BEGIN_ROUND        N1 COLOR_2 N13 COLOR_1 N15 COLOR_2 N19 COLOR_3 ...
  9 deduced
BEGIN_ROUND        N42 COLOR_2 N46 COLOR_0 N93 COLOR_1
  3 deduced
BEGIN_ROUND        N88 COLOR_3
  1 deduced
... 1 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours [67, 84, 34, 49, 59]

colours known before this round:

```
   [0, 1, 3]  (3 = q-1)
```

only colour left: 2

```
so the trace emits  N0 COLOR_2
```

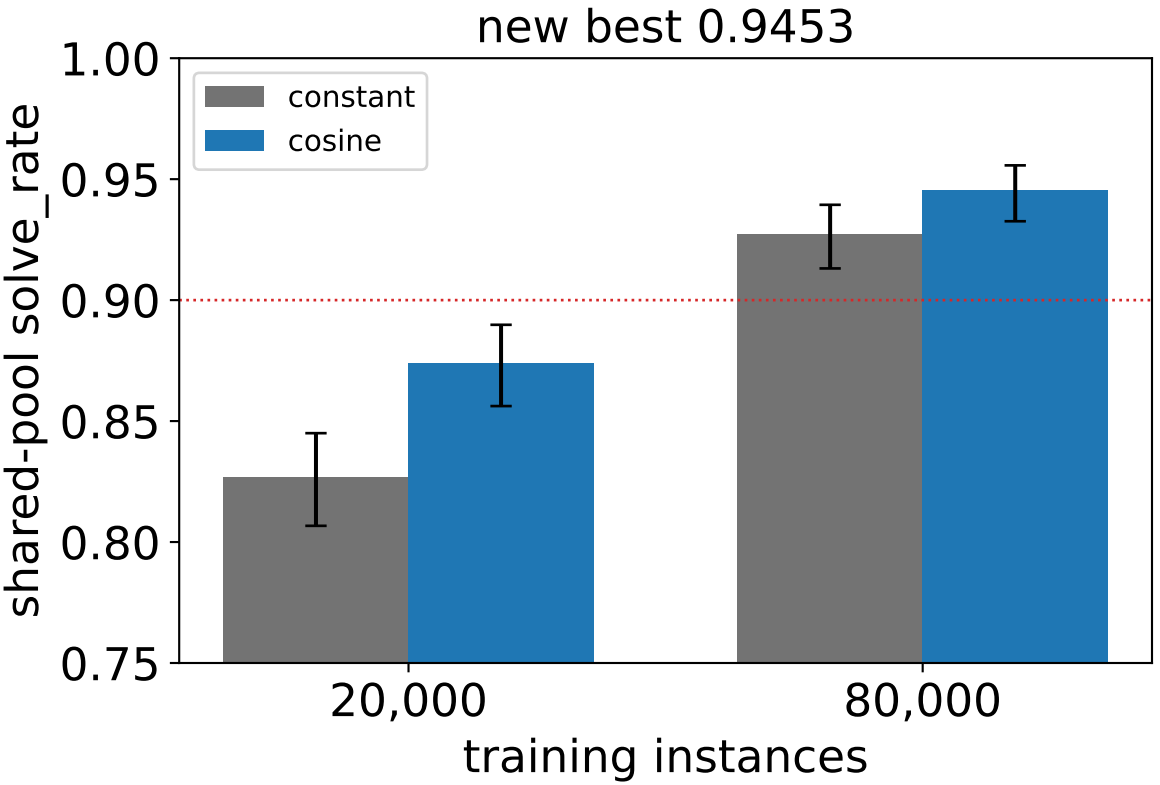
The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

T=5 rounds, the median of 40 draws from this config (range 1-19); seed 50000000002, shuffle_adj=False; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

The 2x2, on one pool

Every cell is one change from its neighbours; every comparison below is paired McNemar exact on the same 1500 instances.



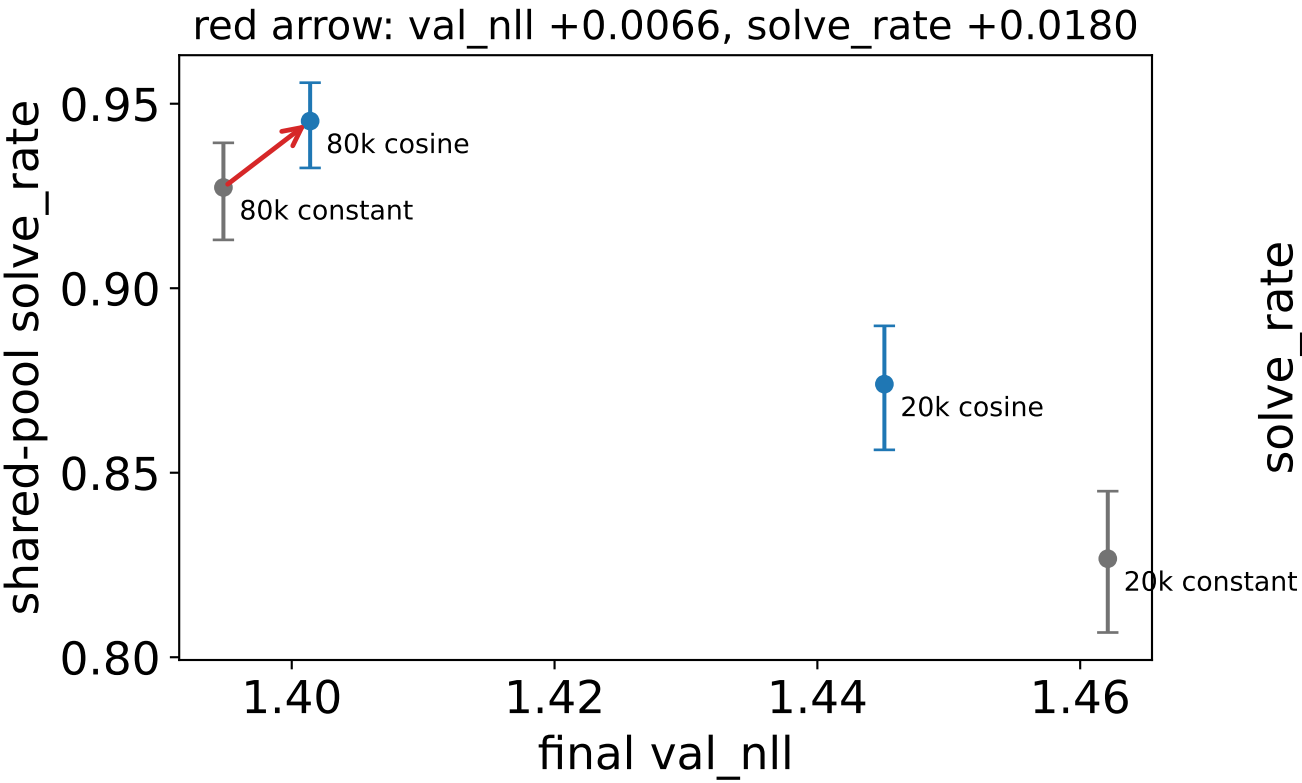
comparison	delta	disc.	p
cosine @ 20000	+0.0473	180-109	3.5e-05
cosine @ 80000	+0.0180	88-61	0.033
4x data @ constant	+0.1007	215-64	3e-20
4x data @ cosine	+0.0713	153-46	1.3e-14

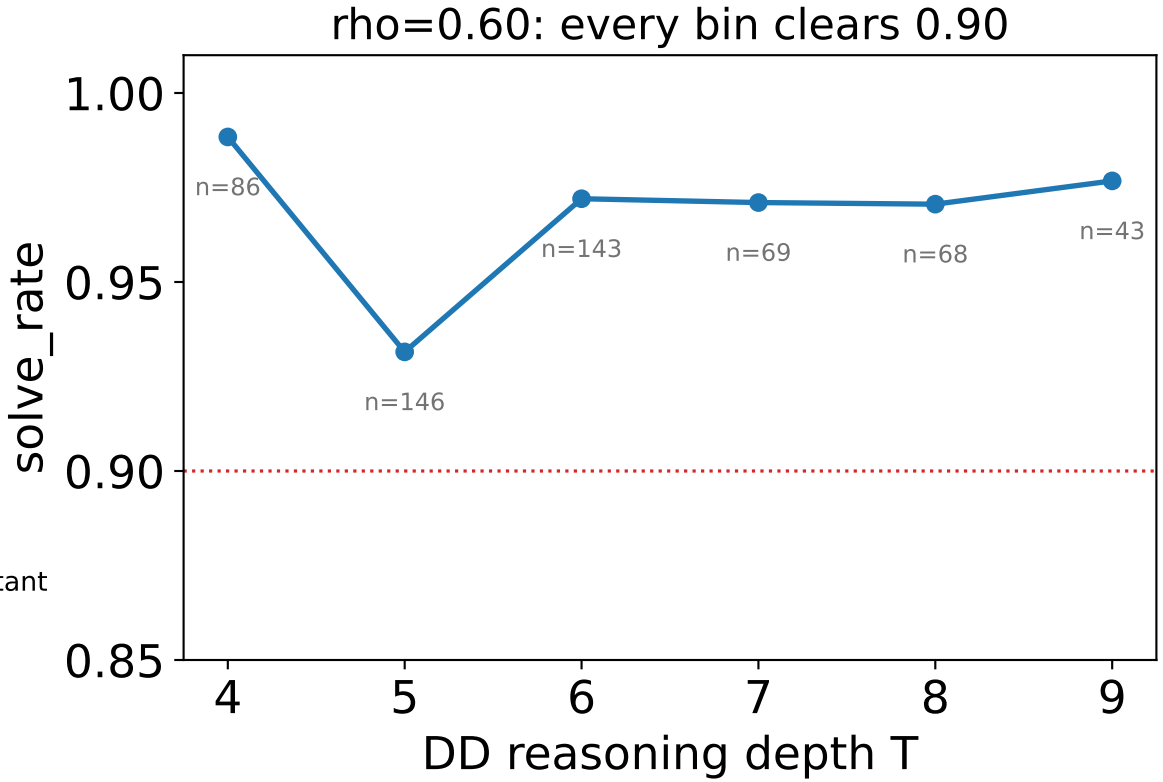
Cosine is worth 2.6x less once the data budget is 4x larger (+0.0473 -> +0.0180), and 4x data is worth less once the schedule is already cosine (+0.1007 -> +0.0713). Stacking both gives +0.1187 against the +0.1480 the single-lever gains predict if they added: an interaction of -0.0293. Diminishing returns, not independence -- which also means neither single-lever number generalises to a different budget.

reference cell for the CSV is 80000/constant (0.9273); the other three deltas are recomputed pairwise from sharedpool_n1500_per_instance.npz

val_nll went the wrong way

The new best has a HIGHER val_nll than the cell it beats, and beats it anyway. Finding 4, now with one change isolated.





Final val_nll 1.4014 (new best) vs 1.3948 (cell B) -- worse by 0.0066 -- while solve_rate is better by +0.0180 at p = 0.033. The two runs differ in exactly one thing, the lr schedule, so this is not a confounded cross-model comparison: it is the cleanest counterexample yet to reading val_nll as progress. The train-val gap is +0.0221 against cell B's +0.0239, i.e. unchanged -- and per finding 16 that quantity is not a fit diagnostic anyway.

depth bins are 1500 fresh instances at rho=0.60, realized density 0.59375 in every bin; weakest bin is T=5 at 0.932 (n=146)

Pre-registrations, as resolved

Written in the sbatch header before the job ran. Three of four held.

(a)	It beats cell B's 0.9273 paired at p < 0.05	**CONFIRMED**	+0.0180, 88-61, p = 0.033. New best at N=96.
(b)	The gain is SMALLER than cosine's +0.0473 at 20000 (falsifier: a gain >= +0.0473)	**CONFIRMED**	+0.0180, i.e. 2.6x smaller. Flagged as low confidence because it rests on the same train/val reasoning that was wrong by 2.1x in finding 15 -- this time it held.
(c)	Every rho=0.60 bin still clears 0.90	**CONFIRMED**	0.988 / 0.932 / 0.972 / 0.971 / 0.971 / 0.977 over T = 4..9. Cell B's weakest bin was T=9 at 0.907.
(d)	The train-val gap widens past +0.0239	**FALSIFIED**	+0.0221 -- narrower by 0.0018, i.e. unchanged. The prediction should not have been made: finding 16 had already shown this quantity is not a fit diagnostic.

(b) is the first time the train/val-gap style of reasoning has predicted correctly in this project; it was still the weakest-grounded of the four

What this is worth

IT DOES SHOW

A new best at N=96 on the metric of record, with all four 2x2 cells paired on one pool.

That the two levers are sub-additive, so single-lever gains measured at one budget do not transfer to another.

IT DOES NOT SHOW

Anything about reasoning. These four checkpoints were trained on the LEAKED token order (finding 16), so every number here is a DECODING score: the N=96 record collapses to 0.0375 under adj-permutation.

One seed per cell. The depth bins are from one draw, n = 43..146 per bin.

WHAT I WOULD DO NEXT. The N=96 optimisation question is now answered well enough; the open question worth compute is the same recipe trained leak-free, where N=24 already costs 0.996 -> 0.854 and reasoning depth finally bites (findings 16-17). A leak-free N=96 run is the honest version of this page.